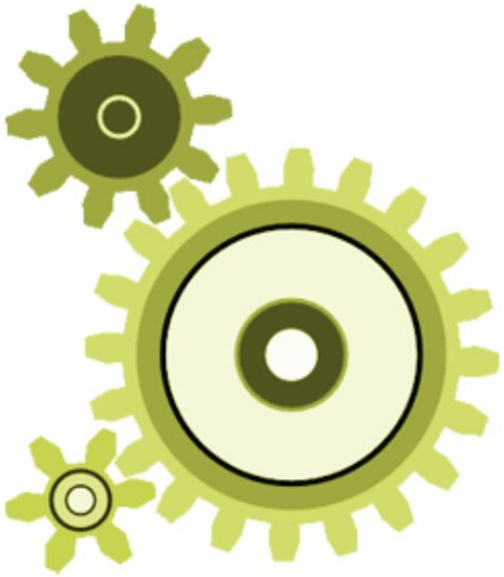
Internal: Augment



Problem Statement:

- Augment existing analysis with a general purpose Tree -> Nextstrain pipeline

Methods:

- Nextflow
- Nextstrain

Status:

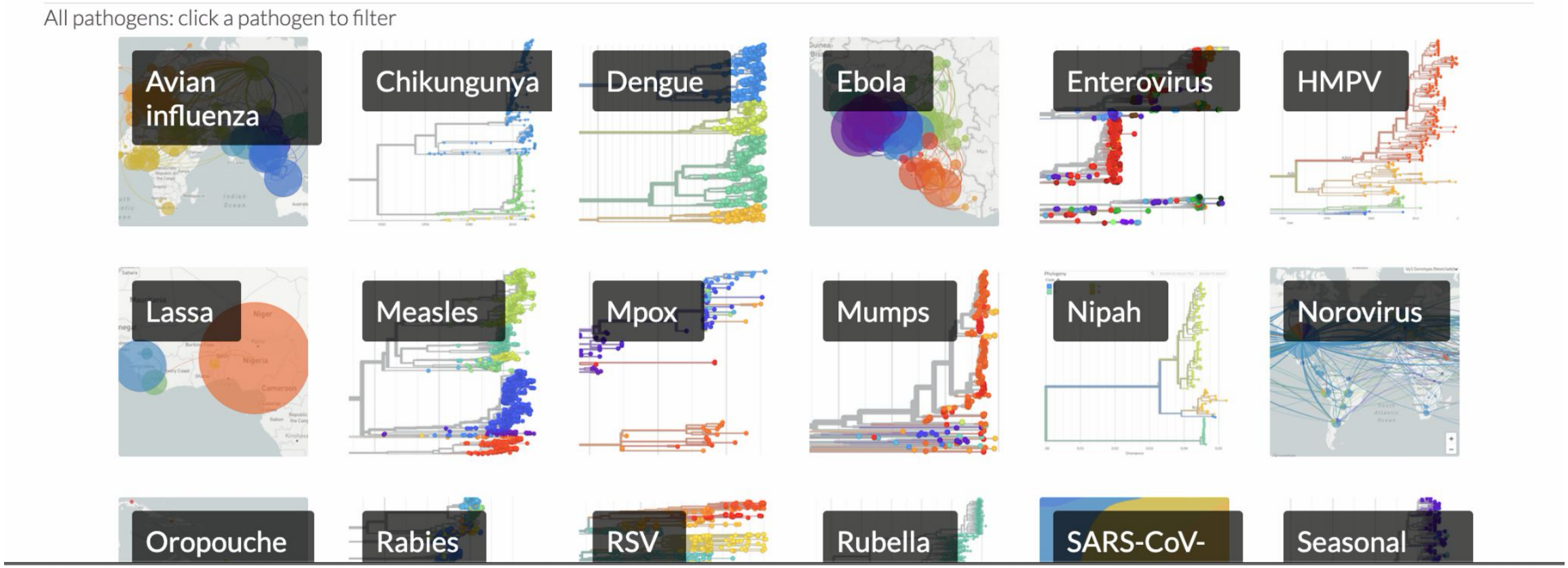
- Initial Draft: https://github.com/j23414/augment

Potential Challenges:

- Defining genbank files
- Defining complex alignments
- Defining complex IDs for tanglegrams
- …

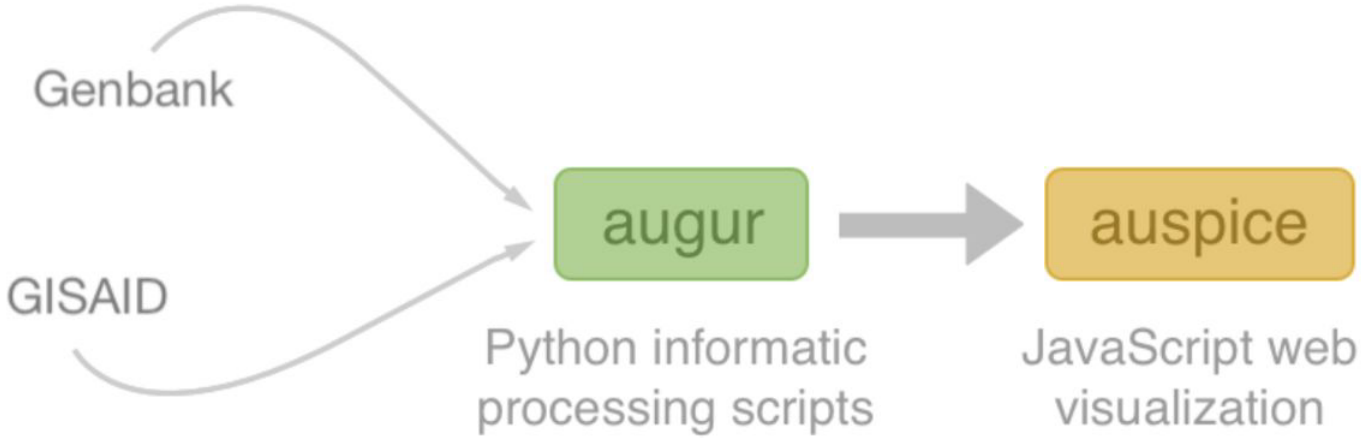
Nextstrain

Real-time tracking of pathogen evolution



Nextstrain architecture

Central aims: (1) rapid and flexible bioinformatic workflows, (2) interactive visualization and (3) always up-to-date analyses at nextstrain.org



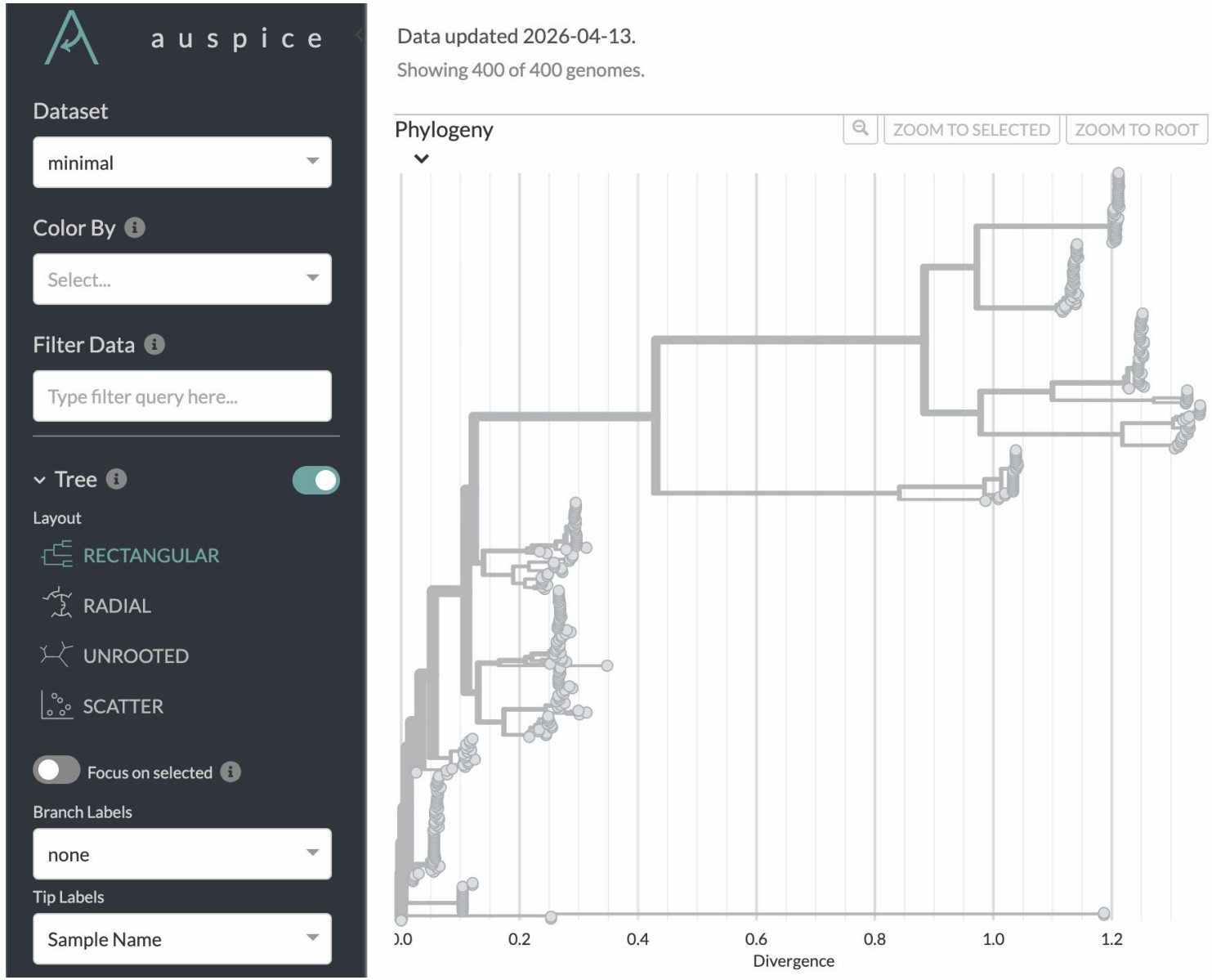
All code open source at github.com/nextstrain

Hadfield et al. 2018. Bioinformatics.

Just the tree

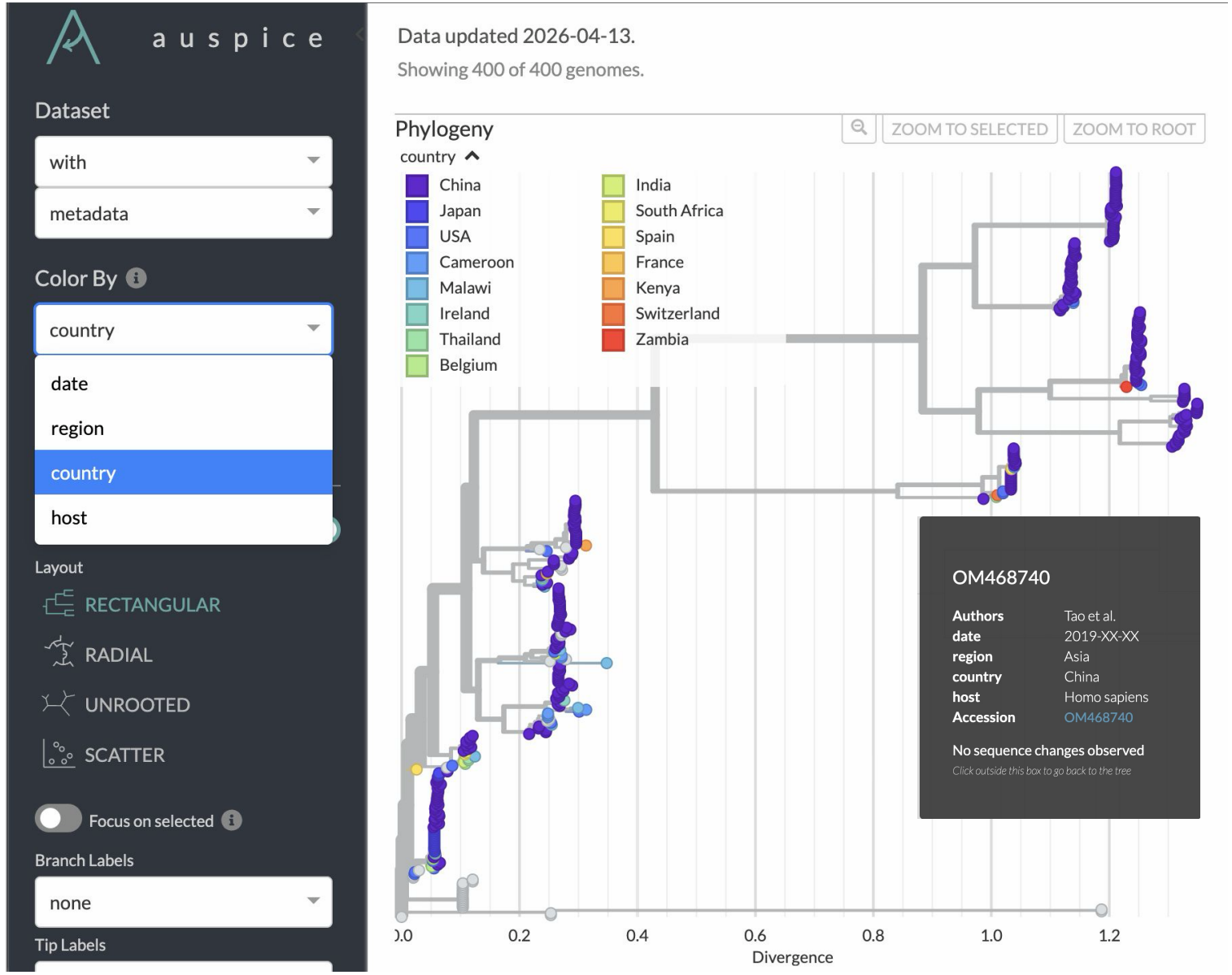
```
nextflow run j23414/augment \
  --newick tree.nwk

nextstrain view results/export
```



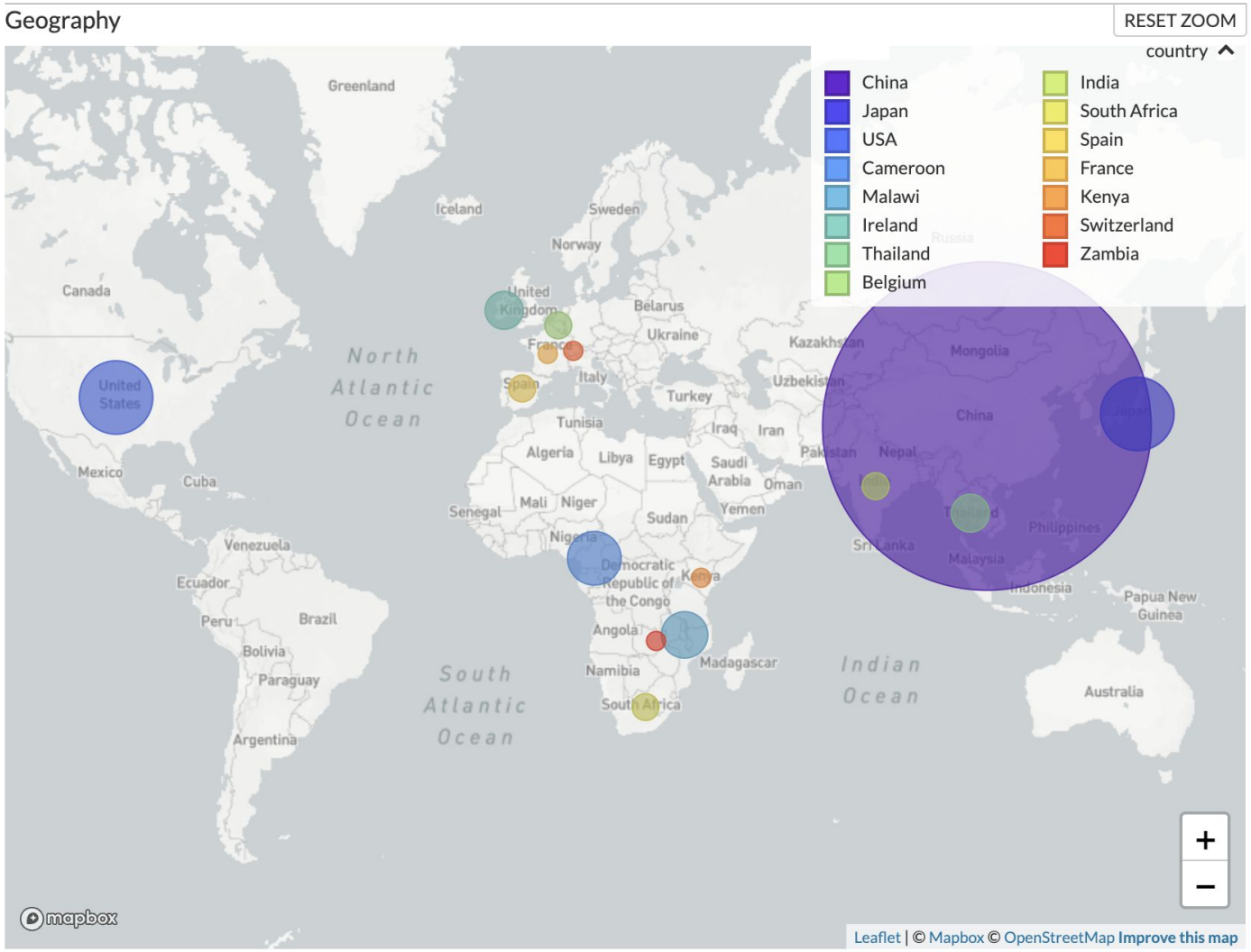
Add metadata

```
nextflow run j23414/augment \
  --newick tree.nwk \
  --metadata metadata.tsv \
  --metadata_id_columns accession \
  --metadata_annotate \
    "date region country host"
```



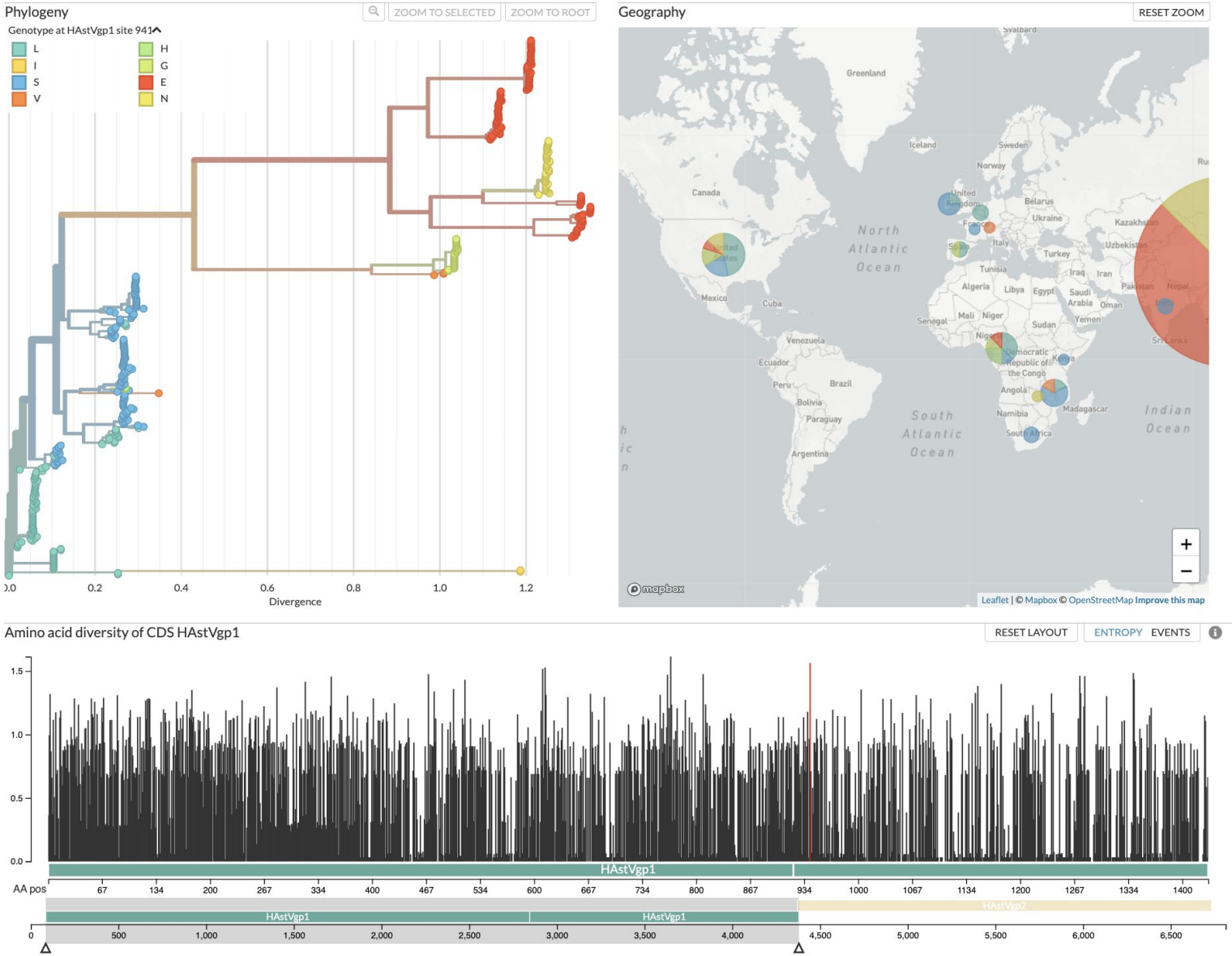
Add geo-resolution to get map

```
nextflow run j23414/augment \
  --newick tree.nwk \
  --metadata metadata.tsv \
  --metadata_id_columns accession \
  --metadata_annotate \
    "date region country host" \
  --export_params \
    "--geo-resolutions country"
```



Mutations

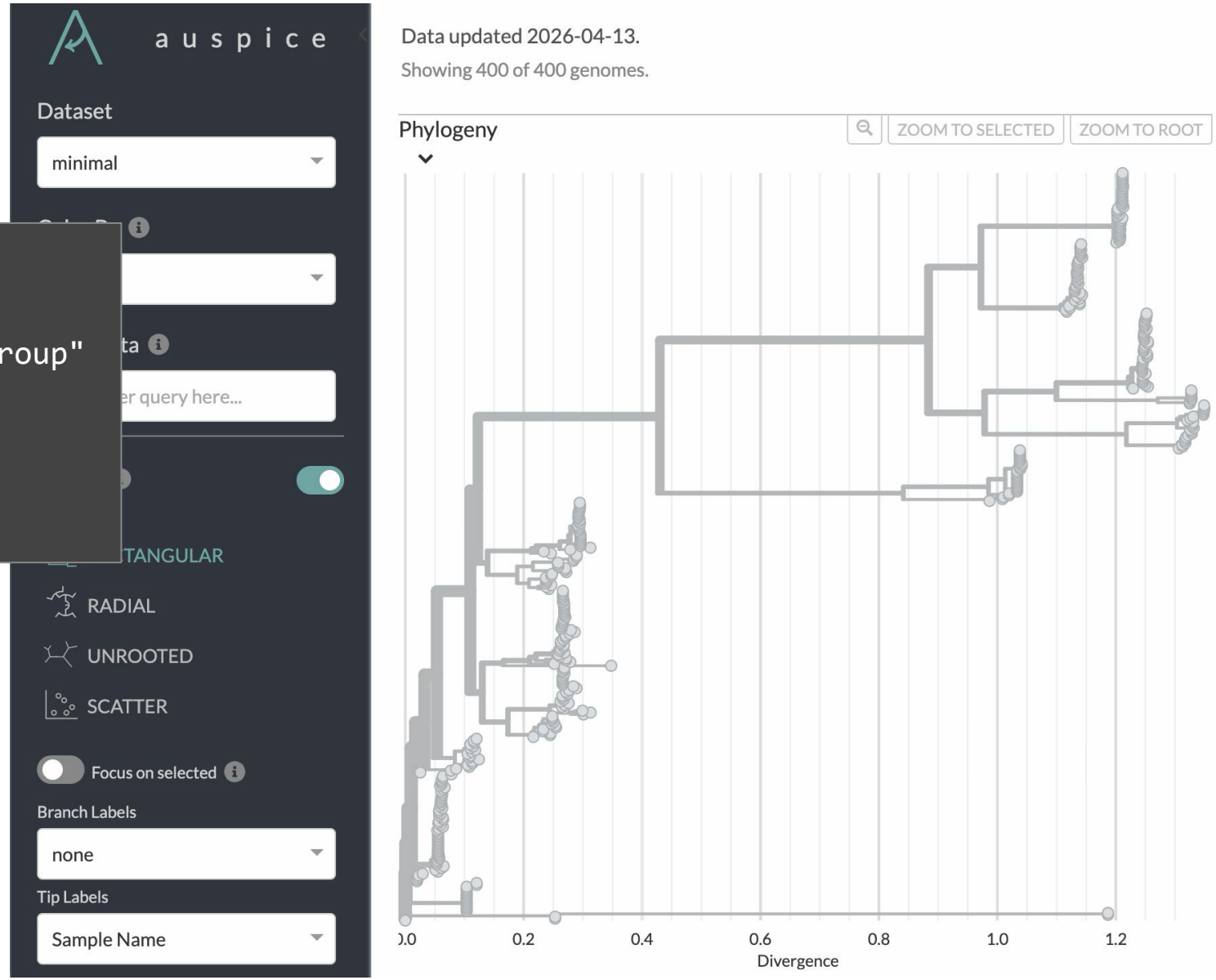
```
nextflow run j23414/augment \
  --newick tree.nwk \
  --metadata metadata.tsv \
  --metadata_id_columns accession \
  --metadata_annotate \
    "date region country host" \
  --export_params \
    "--geo-resolutions country" \
  --alignment alignment.fasta \
  --reference_gb reference.gb \
  --reference_fasta reference.fasta
```



Reroot the tree

```
nextflow run j23414/augment \
  --newick tree.nwk \
  --refine_params \
    "--root OUTGROUP_ID --remove-outgroup"

nextstrain view results/export
```



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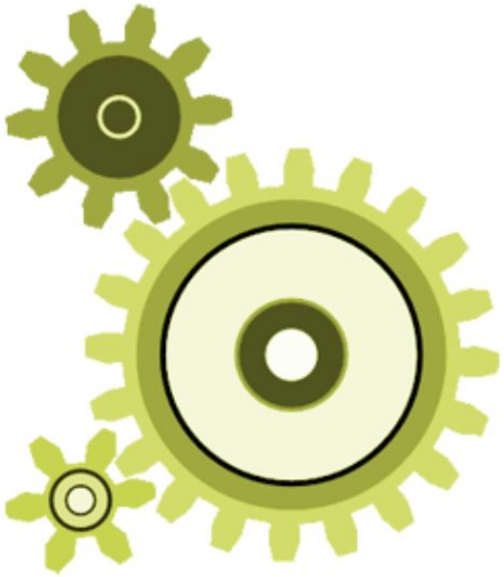

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